

Josiah Holder

Mechanical Engineer & Hands-On Builder

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SUMMARY

Mechanical engineer in Research and Advanced Engineering at Precision Planting, taking products from concept through prototyping and testing. Co-founder of GrainWave Technologies, building wireless control hardware and the software that supports it. Builds things at home when the garage needs something, usually modeling it in CAD first.

EXPERIENCE

Research & Advanced Engineering - *Precision Planting* 2023 - Present

Take products from concept through prototyping, testing, and optimization on small teams of up to five engineers, owning a piece of the work end to end.

Foreman - *Faul Building and Repair* 2021

Ran a crew of up to eight, training and delegating.

High School Intern - *Appareo* 2019

Production data analysis and manufacturing quality checks, running Cpk and Ppk against tolerance and standard deviation.

PROJECTS

PowerSync

Mechatronics - Wireless PTO Control

Wireless remote for tractor PTO control, designed and built end to end, remote and controller both.

GrainWave Partners

Software - Dealer Ordering Portal

System GrainWave's dealers use to order parts, replacing a shared spreadsheet, plus parts catalog and returns.

Shop Storage Cabinet

Hardware - Fusion 360 and Plywood

2x3x6 ft cabinet planned in CAD to fit filament, motors, boards, and CNC bits with no wasted space.

GrainWaveLink

Software - Firmware Delivery

Updates every PowerSync device over Bluetooth, no cable or app store, works offline once downloaded.

Gladiator

Mechatronics - Combat Robot

Wedge-style combat robot for Robot Smashing League, designed to get under opponents and flip them.

Coffee Cart

Hardware - Modeled and Built

Beverage cart for church use, sized to a tight footprint and modeled first before any wood was cut.

EDUCATION

B.S. Mechanical Engineering - *North Dakota State University*

December 2024